

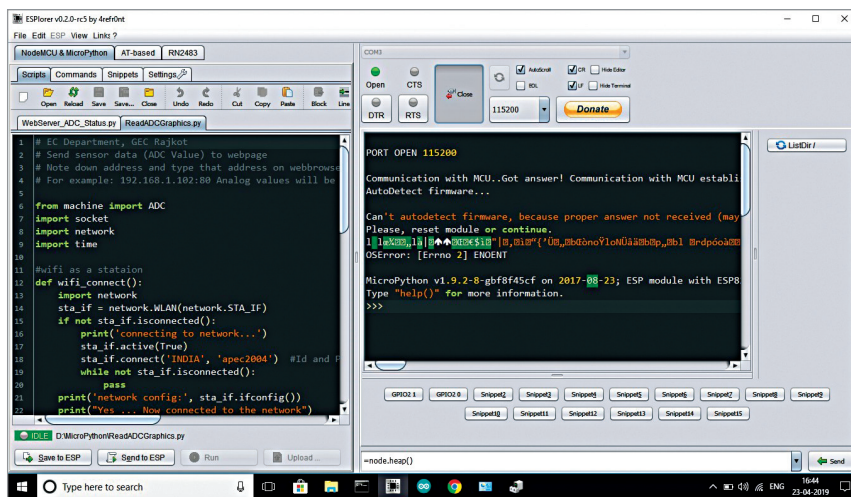


Fig. 4: ESPlorer screen when reset button is pressed

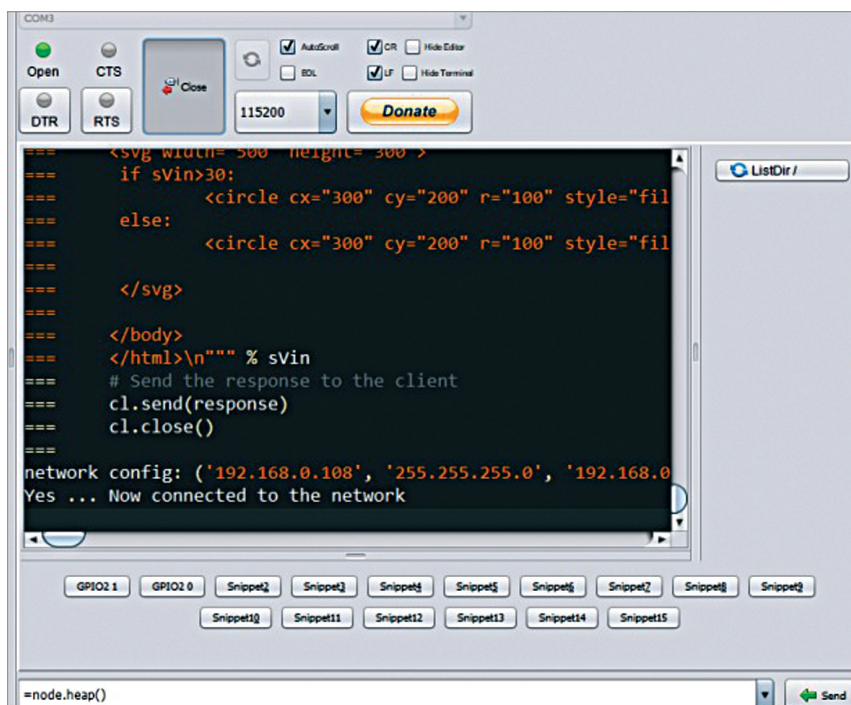


Fig. 5: IP address displayed on ESPlorer

shown in Fig. 1. The circuit diagram to measure and monitor temperature remotely is shown in Fig. 2. Parts list for the circuit includes NodeMCU ESP8266-12 board and an LM35 sensor or digital/analogue I/O board.

NodeMCU is used as a web server that measures temperature and broadcasts the sensor data on the Internet browser.

Software

The NodeMCU board can be programmed with Arduino IDE or MicroPython firmware.

Install Arduino IDE and NodeMCU support using Boards Manager. This installation process using Arduino was published in 'Control With Authentication for Up to Four Devices' DIY article in April 2019 issue of EFY magazine, which is now available at <https://www.electronicshobby.com/electronics-projects/authentication-device-control-circuit>

MicroPython allows you to run miniature version of Python 3 on embedded boards and microcontrollers. It is a tiny open source Python programming language interpreter

designed to run on small embedded devices. MicroPython does not support entire Python standard library, so we can call it a sub-set of Python.

Advantages of MicroPython include small footprint (requires less memory), it is near the hardware due to Hardware API, and is a clean and simple Python code to control hardware. Disadvantages include limited standard library functions and execution speed less than of C language. So sometimes it may create problem for real-time signal processing where execution speed requirement is critical.

There are two methods to install MicroPython firmware in NodeMCU. Use any one of the following methods (ESP8266 flasher and ESptool) to upload MicroPython firmware to your NodeMCU.

ESP8266 flasher. Download *nodemcu-flasher* from <https://github.com/nodemcu/nodemcu-flasher>. Download binary file *esp8266-20170823-v1.9.2.bin* (or the latest version). Click ESP8266Flasher in *nodemcu-flasher* unzip folder, select binary file *esp8266-20170823-v1.9.2.bin* firmware, and click on Flash as shown in Fig. 3. This will install the firmware in NodeMCU board.

ESptool command. Use ESptool in command prompt. Following commands are used in Windows 7, 64-bit machine with Python 3. Open command prompt (cmd) and install esptool using following command:

```
>pip install esptool
```

First connect NodeMCU board to your PC and erase it using following command:

```
>esptool.exe --port COM9 --baud 115200 erase_flash
```

Change COM9 port depending on COM port assigned to the NodeMCU after connecting it with your PC/laptop.

Next, download firmware to NodeMCU using following command:

```
> esptool.exe --port COM9 --baud 115200
write_flash --flash_size=detect 0 esp8266-20170823-v1.9.2.bin
```

In above command, give the path of binary file for the firmware (if it is not there in the same directory), use *flash_size=detect* if ROM size is unknown. (Above command can be written as batch file.)

Now, download ESPlorer software from <https://esp8266.ru/esplorer/>